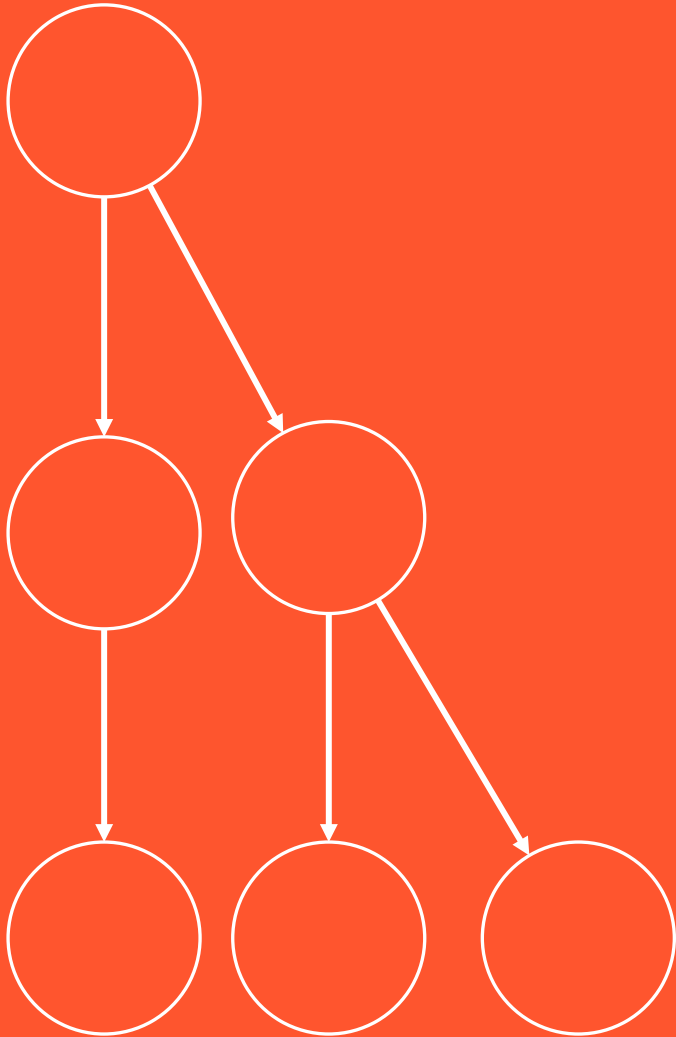




BCOG 100

Course Overview

Lecture 0 — What Is Brain and Cognitive Science?



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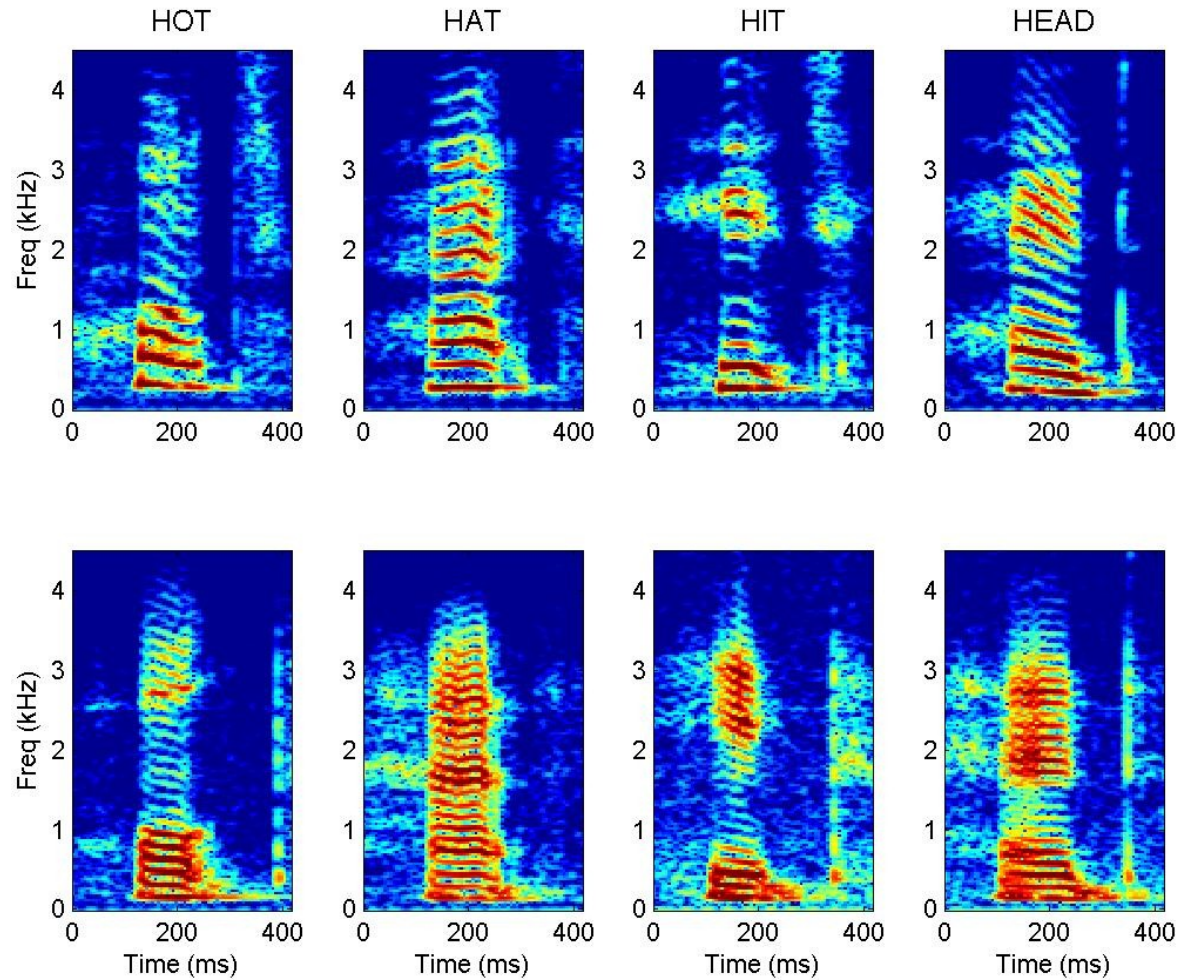
Look At What You Can Do



Look At What You Can Do



Look At What You Can Do



Look At What You Can Do



Getty/indianexpress.com

And This One Is Three Years Old



Minds Built Nothing Like Ours



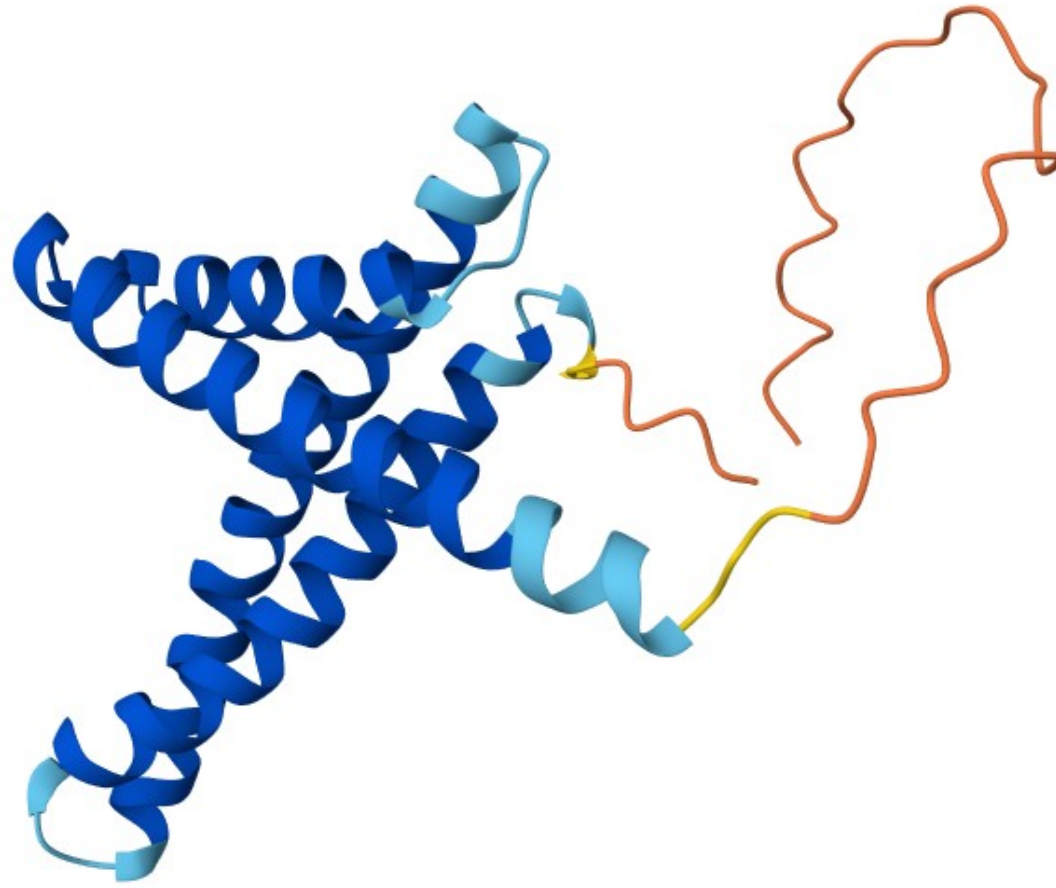
Minds Built Nothing Like Ours



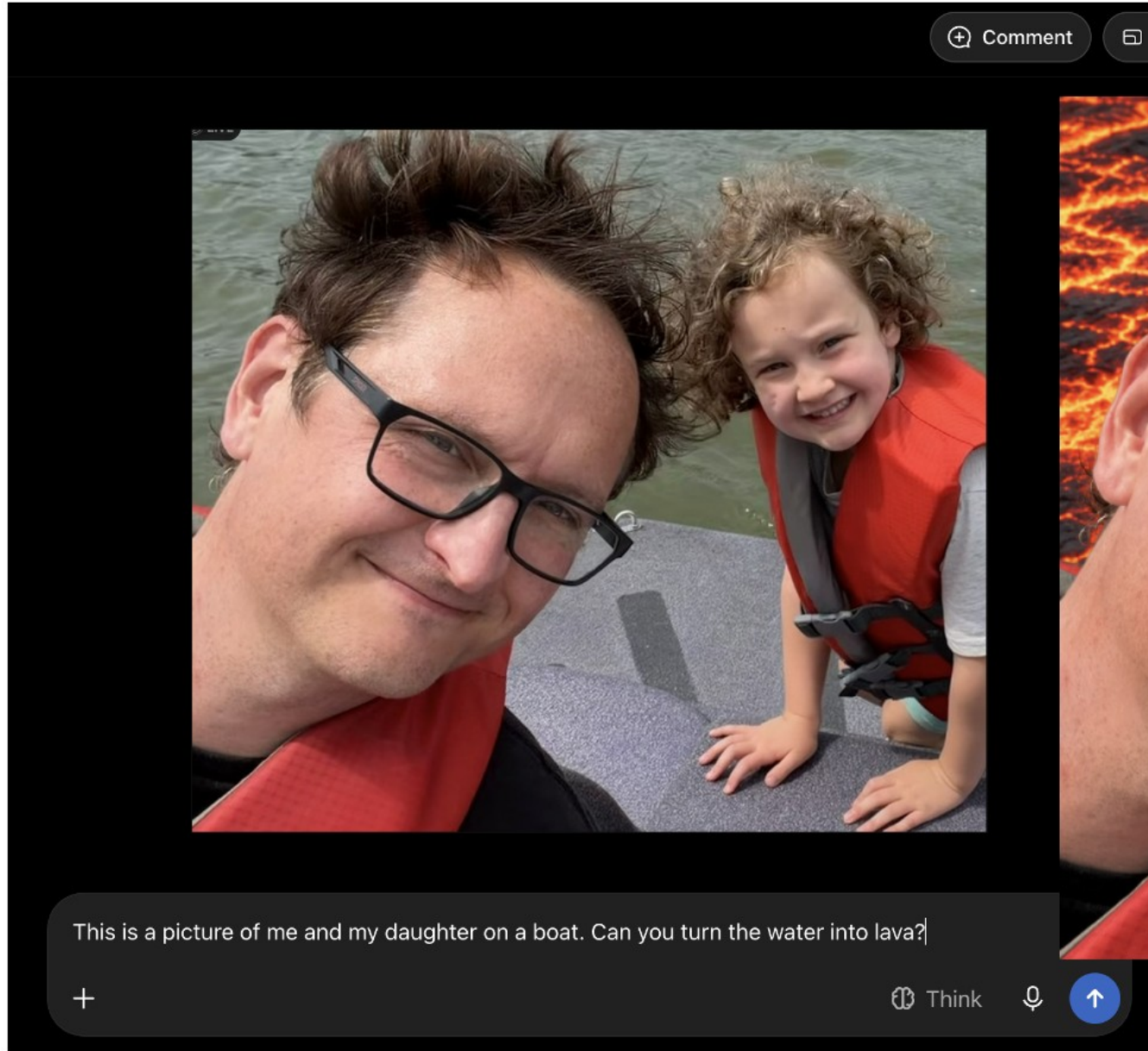
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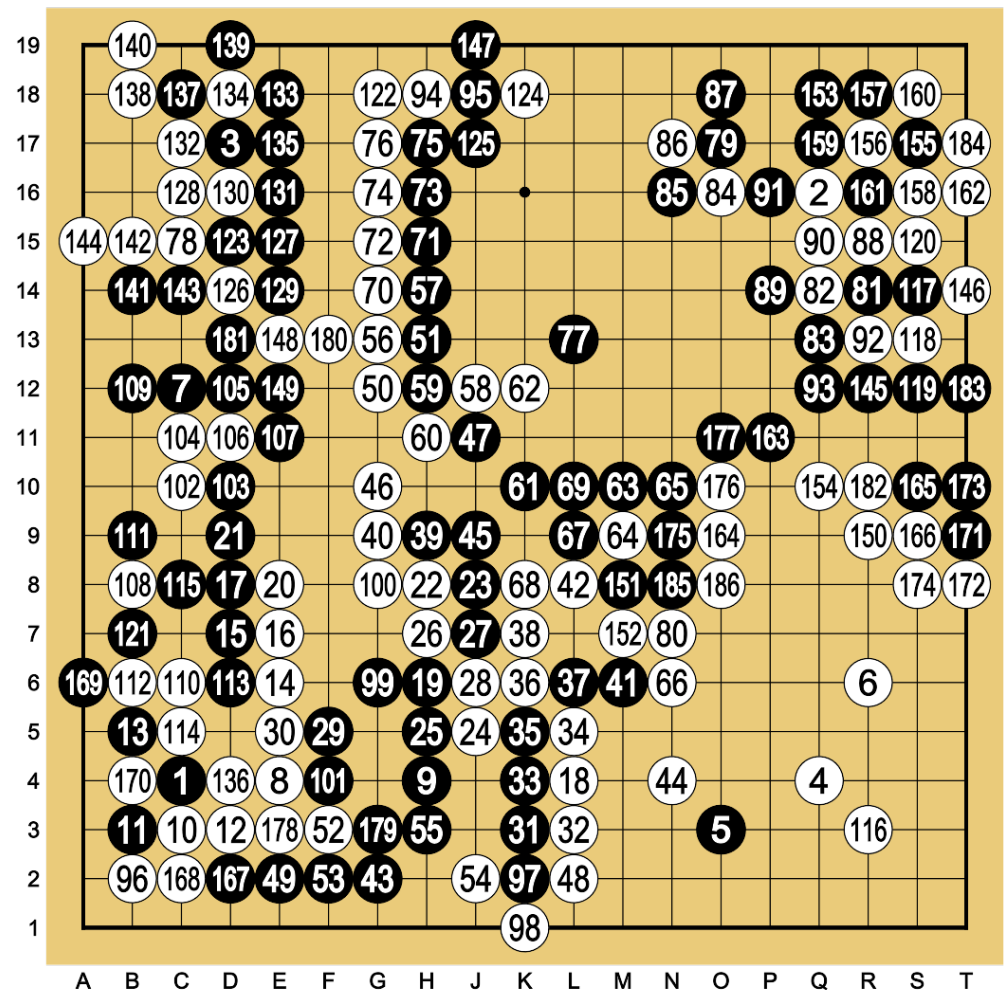
Minds With No Neurons At All



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Minds With No Neurons At All



Minds With No Neurons At All

I am planning on going to see weezer in Chicago at the united center. Can you give me advice about the best section to sit in?

Searched the web

Good news first — the show is confirmed and well-documented. Weezer plays Chicago on Tuesday, September 22, 2026, as part of "The Gathering" tour, with The Shins and Silversun Pickups as supporting acts. Doors-to-stage timing starts at 7:00 PM. Three real bands on one bill means a long night, so seat comfort matters more than it would for a single-act show.

Choose Chicago Shazam

The United Center runs a standard end-stage setup for most concerts. The stage sits at the end near sections 115, 116, and 117, with the floor divided into six sections — 1 through 3 closest to the stage, 4 through 6 behind them. [The Stadiums Guide](#)



Write a message...

+

Opus 5 High ▾



Claude is AI and can make mistakes. Please double-check cited sources.

Twenty Watts

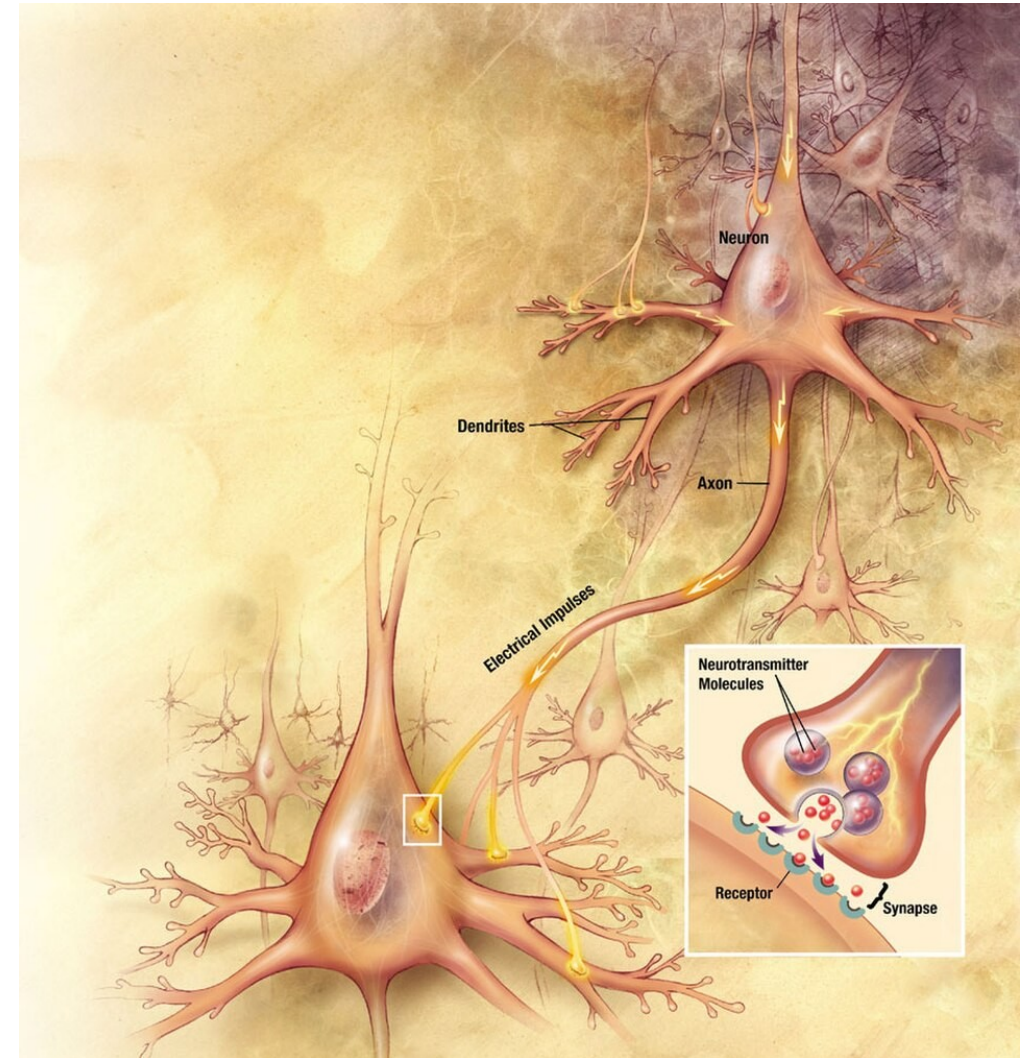
One neuron is a fairly simple machine

You have 86 billion of them, each in conversation with thousands of others

The whole thing runs on about 20 watts — less than a household lightbulb

The machines that fold proteins, play Go, and run chatbots fill warehouses and draw the power of a small town

How do you get all of that out of this?



Twenty Watts

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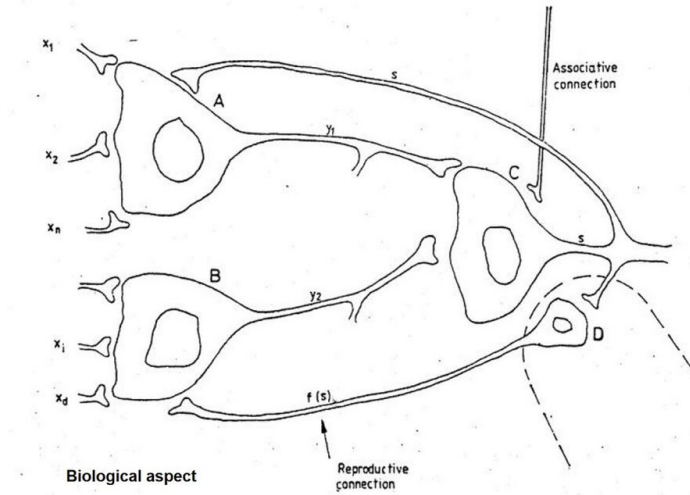
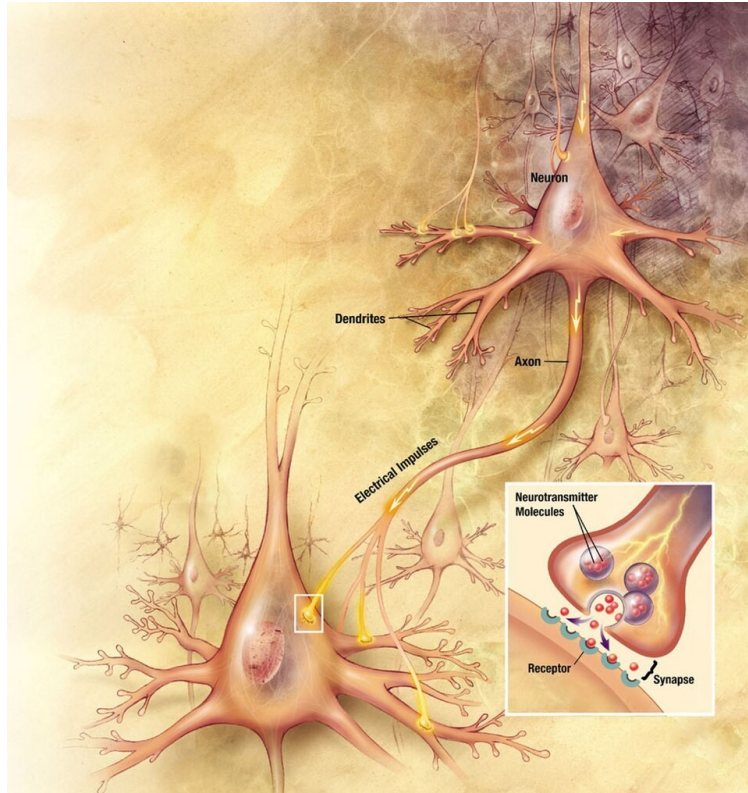
Which of These Is Most Impressive?

Which of the things we just saw is the most impressive?

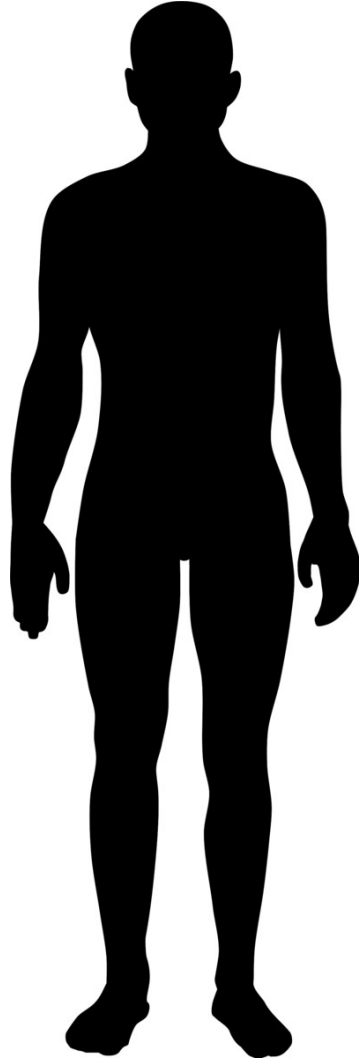
Which one is least like the others?

Which of them, if any, has a mind?

What Could Possibly Explain All of These?



What Could Possibly Explain All of These?



What Could Possibly Explain All of These?



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115-117 · behind stage

113-114

↓

118-119

Write a message...

+ Opus 5 High ▾ 🔊 🔊

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What Is Brain and Cognitive Science?

The study of the mind and the brain, with two commitments:

1. Interdisciplinary — no one field can explain a mind by itself
2. Computational — minds take in information, store, represent, or are affected by the info, and act on it

An ambition more than an accomplished fact

Interdisciplinary BCS

Psychology

Neuroscience

Philosophy

Computer Science and Artificial Intelligence

Anthropology

Linguistics

Education

Ecology and Evolutionary Biology

One Moment, Five Questions

A three-year-old says: "I goed to the store."

Neuroscientist — what has to be in place, physically, for her to produce a form she has never once heard?

Psychologist — did she look up a word, or did she build something and apply it?

AI researcher — what would a system need, in order to make exactly this mistake?

Linguist — "goed" is nowhere in what she heard. What does the error reveal about the grammar she has already built?

Philosopher — is "she has a rule" a claim about what is in her head, or just a tidy description of her behaviour?

Computational BCS

Computational Metaphor

- Information comes in, is held or transformed by the system somehow, and an action comes out
- A common shorthand: the brain is the hardware, the mind is the software
- Two questions follow, and they run through the whole course:
 - how does a mind hold on to information, and how does it act on what it holds

Computational Methods

- Computer-based experiments
- Computer-driven tools like EEG and fMRI
- Complex data analysis and machine learning
- Computational models and simulations

Is the Brain Really a Computer?

The brain has no clean divide between hardware and software

It does not run in the tidy, step-by-step way a laptop does

A mind is bound up with a body and a world, not sealed inside a skull

"Computation" may have stretched so far that it explains nothing

A lens, not a doctrine

Three Levels, Three Questions

What problem is this system solving, and why?

How does it solve that problem — what does it represent, and what processes act on it?

What is it physically made of, and how does that stuff carry the work out?

Not three rival answers. Three different questions.

Where We're Going

We follow the order evolution built things in — and at each step, we compare against machines.

What evolution added

The first neurons

Neural circuits, affect and valence

Learning and plasticity

The vertebrate brain

Pattern recognition

Perception and action

Spatial cognition

Memory, imagination and generative models

Social cognition

Planning and decision-making

Language

Symbolic cognition

Culture

The problem it solved

Sense and act faster than chemistry alone allows

Work out whether a thing is good or bad, and do something about it

Change with experience, instead of being finished at birth

Run many systems at once, in one body

Find the same thing again, having never seen it the same way twice

Act on a world you can only ever sample

Know where you are, and get back

Use the past, and run a future before living it

Predict the most complicated thing in the environment: other minds

Choose between futures that have not happened

Put a thought into someone else's head

Think about what is not present, and what is not real

Inherit what others worked out, without working it out yourself

What a Week Looks Like

The reading is posted before the week it covers begins

Three lectures

One lab

A five-minute quiz at the start of most classes — starting in week 2, and always about the week before

Practice questions, ungraded and unlimited, on Canvas

How You're Graded

Daily in-class quizzes — 136 points, 34% (40 given, 34 counted)

Lab assignments — 120 points, 30% (14 given, 12 counted)

Final exam — 144 points, 36%

Total — 400 points

Extra credit: research participation (SONA) up to 3%, course notes up to 3%

Generous drops, which is why there are no make-ups

Why It's Built This Way

Practice questions on Canvas: ungraded, unlimited, with explanations for every answer

The daily quiz is on paper, closed book, drawn from those same practice questions

Quizzes start in week 2, and each one covers the week before — you always get a week with the material first

The final is drawn from them too

No midterm — the daily quizzes already spread the observations out

How to Do Well

Come to class. Do the reading before the week starts. Work the practice questions.
Take notes with a real method — SQ3R, Cornell Notes — described in a document on Canvas.
Ask questions, and come to office hours

Cues→

Cornell Style Notes

08/24/26	Lecture 0-1	What is BCS?
2 commitments	BCS = study of mind + brain (1) interdisciplinary (2) computational ⚠ an ambition, not a finished field	
How do the fields differ?	8 disciplines — differ by the QUESTION, not subject ex: 3-yr-old says "I <u>goed</u> " → 5 questions not a ranking → integration	
"Computational" — 2 senses?	Computation = metaphor (info in → held/transformed → action out) AND method (models, fMRI, ML)	
Objections?	But: no clean <u>hw/sw</u> divide; not step-by-step; mind bound up w/ body + world; word too elastic → a LENS, not a doctrine	

←Notes

Summary

How to Do Well

Beware the illusion of understanding — following an explanation is not the same as being able to produce it

The fix is not rereading. It is closing the thing and making yourself produce the answer, badly, before you check

Using AI in This Course

Not this. Do not have a model produce answers, text, code, or figures you hand in as your own work. That covers every lab in this course

Does ChatGPT make people more dumb, because they outsource all their thinking?

That's a fair and important question—and it's being actively debated.

The concern:

Yes, **there's a real risk** that if people **over-rely** on ChatGPT or similar tools, they may:

- **Stop practicing critical thinking**, problem-solving, or independent writing.
- Default to **copy-paste answers** rather than understanding concepts deeply.
- Accept information without question, especially if it's packaged well.

This kind of passive use **can erode cognitive skills** over time—just like using a calculator for every little thing might weaken your mental math.

Using AI in This Course

This, and I encourage it. Have it re-explain a passage you have already read • quiz you, answer withheld until you commit • decode notation • argue with reasoning you have already worked out

If you finish and could not now do the thing yourself, it did not help you

Each lab report ends with a sentence on whether you used AI and how. It does not affect your grade

But it doesn't have to go that way:

Used intentionally, ChatGPT can actually **make you smarter** by:

- Acting as a **sounding board** to clarify your thoughts.
- Offering **different perspectives** or ideas you hadn't considered.
- Helping you **learn faster** by breaking down complex ideas.
- Letting you test and refine your own reasoning by comparing it to an alternative.

It's kind of like Google, Wikipedia, or even books: they **can be tools for learning**, or they can be shortcuts that undermine it—it depends **how** you use them.

So, what's the difference between smart use and dumb use?

Smart Use	Dumb Use
Asking for explanations, then reflecting or researching further	Blindly copying answers
Using ChatGPT to test your understanding or debate ideas	Letting it do all the work
Treating it like a collaborator or tutor	Treating it like a crutch

Why the Line Is Where It Is

These tools **raise the ceiling** for someone who already has the skill, and **lower the floor** for someone who does not — for the same reason. The model does part of the reasoning

An expert is checking the output. A novice is trusting it, and usually cannot tell which they are doing

Why the Line Is Where It Is

Experienced developers, using AI on their own job:

- expected to be **24% faster**
- afterwards believed they *had been* **20% faster**
- measured against the clock: **19% slower**

They were not slightly off. They were wrong about the direction

Take Care of Yourself

Sleep, eat, move, keep a schedule you can actually keep

Wellbeing, writing, and disability resources are linked on the syllabus

Asking early works far better than asking late

Before Next Class

Read Chapter 1 — Mind and Brain

The question we open with: is your mind nothing but your brain?

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